GGG-H-575
November 10, 1966
SUPERSEDING
MIL-H-795A
January 12, 1962

FEDERAL SPECIFICATION

HOLDERS; SHEET METAL, HOLE AND EDGE GRIP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sheet-metal holders for clamping sheets in place during fabrication.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and sizes.* Sheet-metal holders shall be of the following types, classes and sizes, as specified (see 6.2):

Type I—Hole grip.
  Class 2—Forceps-operated, styles A, D, and E (fig. 1).
    Size—As listed in table II.
Type II—Edge grip.
  Class 1—Lever-operated (fig. 2).
    Size—As listed in table III.
  Class 2—Forceps-operated (fig. 3).
    Size—As listed in table IV.
Type III—Hole grip.
  Class 4—Hand-operated—screw type, styles G, H, and J (fig. 4).
    Size—As listed in table V.

2. APPLICABLE DOCUMENTS

2.1 The folllowing specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:
GGG-F-570—Forceps, Sheet Holder.

Federal Standard:
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-H-15424—Hand Tools; Packaging of.

Military Standards:
MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5120

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3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of holders which are otherwise in accordance with the requirements of this specification.

3.2 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the best quality used for this purpose in commercial practice.

3.3 Design and construction. Sheet holders covered by this specification shall be of the spring-tension type. Type I, hole-grip holders, shall have two moving prongs operating with a fixed pin in such a manner that the prongs are expanded by spring tension in order to insure proper alignment. When the holder is inserted into corresponding holes in two or more sheets, the sheets shall be held together securely. The type II, edge-grip holder, shall have two parallel jaws which clamp the edges of the sheets together by means of spring tension. The type III, hole grip, shall be a hand-operated wingnut, all purpose fastener for use with heavier materials where more pressure is required.

3.4 Springs. The springs shall be of good quality steel and properly heat treated to withstand the imposed loads. The force exerted by the spring shall be as specified in table I.

3.5 Safety. The sheet holders shall be so constructed that they will not spring apart with force and cause injury to operator due to possible failure of any part of holder.

3.5.1 *Swaging.* The method of a swaging operation in the final assembly of type I, class 2 forceps-operated holder, shall be of such construction that constant use of the sheet holder shall not allow the swaged part to cut through the spring-retaining cup during normal usage and allow disassembly of the sheet holder. Holders shall meet safety test requirements of 4.4.3.

3.6 Marking. Unless otherwise specified, each holder shall be permanently and legibly marked with the manufacturer's name or trademark.

3.7 Finish. The finish shall be as specified in tables II through V.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

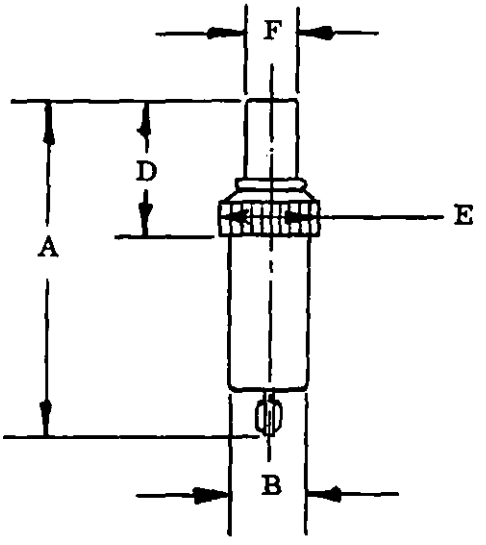
3.9 Type I, hole-grip sheet holder. Type I, hold-grip sheet holders, shall be of the spring-tension, fully enclosed safety type.

3.9.1 *Class 2, forceps-operated.* The class 2, hole-grip sheet holder, shall be so constructed that it is capable of being operated by means of forceps conforming to GGG-F-570. The holder shall be substantially as shown on figure 1, and shall comply with the requirements of table II. The style of holder to be furnished shall be at the option of the contractor. Holders shall be furnished with either a 0 to 1/4 or 0 to 1/2 inch material-holding capacity as specified (see 6.2).

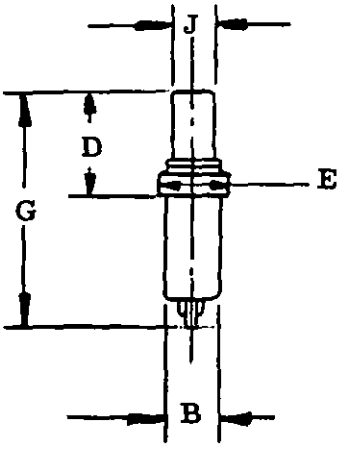
TABLE I. *Requirement for spring tension*

Type	Class	Style	Thickness material clamped	Clamping force (min.)
			Inch	*Pound*
I	2	A, D	1/4	50
I	2	E	1/2	50
II	2	.....	1/2	50
II	2	.....	3/4	50

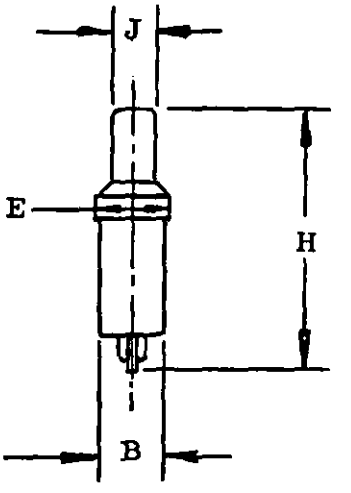
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STYLE A



STYLE D



STYLE E

FIGURE 1. Type I, class 2, hole grip forceps-operated.

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TABLE II. *Detail requirements for type I, cla ss 2, hole grip, forceps-operated*

Nominal rivet size	Hole size standard to oversize	Dimensions A (max.)	B (max.)	C (min.)	D (max.)	E (max.)	F (max.)	Finish, color	G (max.)	H (max.)	J (max.)
Inch	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*		*Inch*	*Inch*	*Inch*
1/16	0.062 to 0.078	2-1/8	0.410	...	15/16	1/2	1/4	Black	1-5/8	...	3-/8
3/32	.093 to .108	2-1/8	.410	5/16	15/16	1/2	1/4	Cadmium	1-5/8	2-7/8	3-/8
1/8	.125 to .140	2-1/8	.410	5/16	15/16	1/2	1/4	Copper	1-5/8	2-7/8	3-/8
5/32	.156 to .171	2-1/8	.410	5/16	15/16	1/2	1/4	Black	1-5/8	2-7/8	3-/8
3/16	.187 to .202	2-1/8	.410	5/16	15/16	1/2	1/4	Brass	1-5/8	2-7/8	3-/8
7/32	.218 to .230	2-1/4	.410	5/16	15/16	1/2	1/4	Cadmium	...	2-7/8	3-/8
1/4	.250 to .265	2-1/4	.410	5/16	15/16	1/2	1/4	Copper	...	2-7/8	3-/8
9/32	.281	3-1/4	.625	5/16	13/16	1/2	5/16	Brass	...	...	...
11/32	.343	3-1/4	.625	5/16	13/16	1/2	5/16	Brass	...	...	...

3.10 Type II, edge-grip sheet holder.

3.10.1 *Class 1, lever-operated.* The class 1, edge-grip sheet holder, shall be operated by means of a hand lever which is a part of the holder. The holder shall be substantially as shown on figure 2, and shall comply with the applicable requirements of table III. The gripping surface of the jaws shall be made of rubber or asbestos as specified (see 6.2).

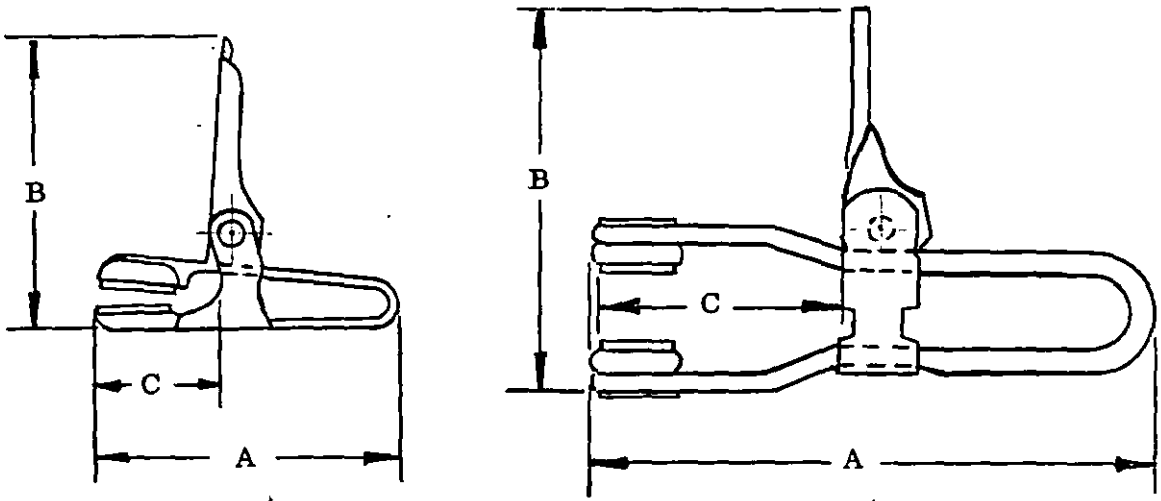


FIGURE 2. Type II, class 1, edge grip, lever-operated.

TABLE III. *Detail requirements for type II, class 1, edge grip, lever-operated*

Size	Jaw opening range	Dimensions A	B	C	Finish, color
Inch	*Inch*	*Inches*	*Inches*	*Inches*	
1/8	0 to 5/32	2	2	3/4	Cadmium
3/8	1/8 to 3/8	3-3/4	3-3/4	1-1/2	Cadmium

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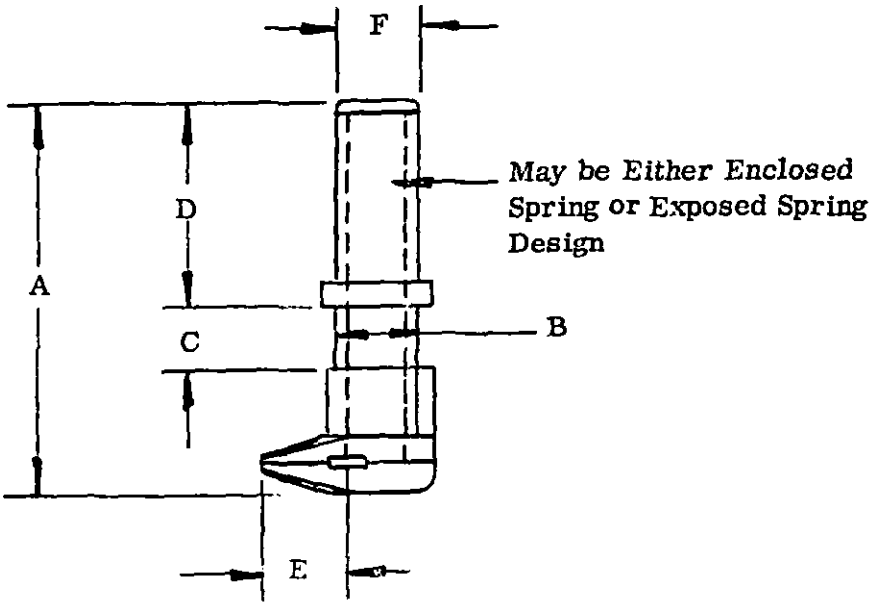


FIGURE 3. Type II, class 2, edge grip, forceps-operated.

TABLE IV. *Detail requirements for type II, class 2, edge grip, forceps-operated*

Jaw opening range	Dimensions						Finish (color)
	A (max.)	B (max.)	C (min.)	D (max.)	E ±1/16	F (max.)	
Inch	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	
0—1/2	2-5/16	0.410	5/16	1	1/2	3/8	Copper or black
0—3/4	2-9/16	.410	5/16	1	1	3/8	Brass or black
Adjustable from 0 to 7/8	3	.410	5/16	1	1	3/8	Copper or black

3.10.2 *Class 2, Forceps-operated.* The class 2, edge-grip sheet holder, shall be capable of being operated by means of forceps conforming to GGG-F-570. The holder shall be substantially as shown on figure 3, and shall comply with the requirements of table IV.

3.11 Type III, hole-grip sheet holder.

3.11.1 *Class 4, hand-operated screw type.* The class 4, hole-grip sheet holder, shall be so constructed that it is capable of being operated by means of a wingnut. The holder shall be substantially as shown in figure 4, and shall comply with the requirements of table V.

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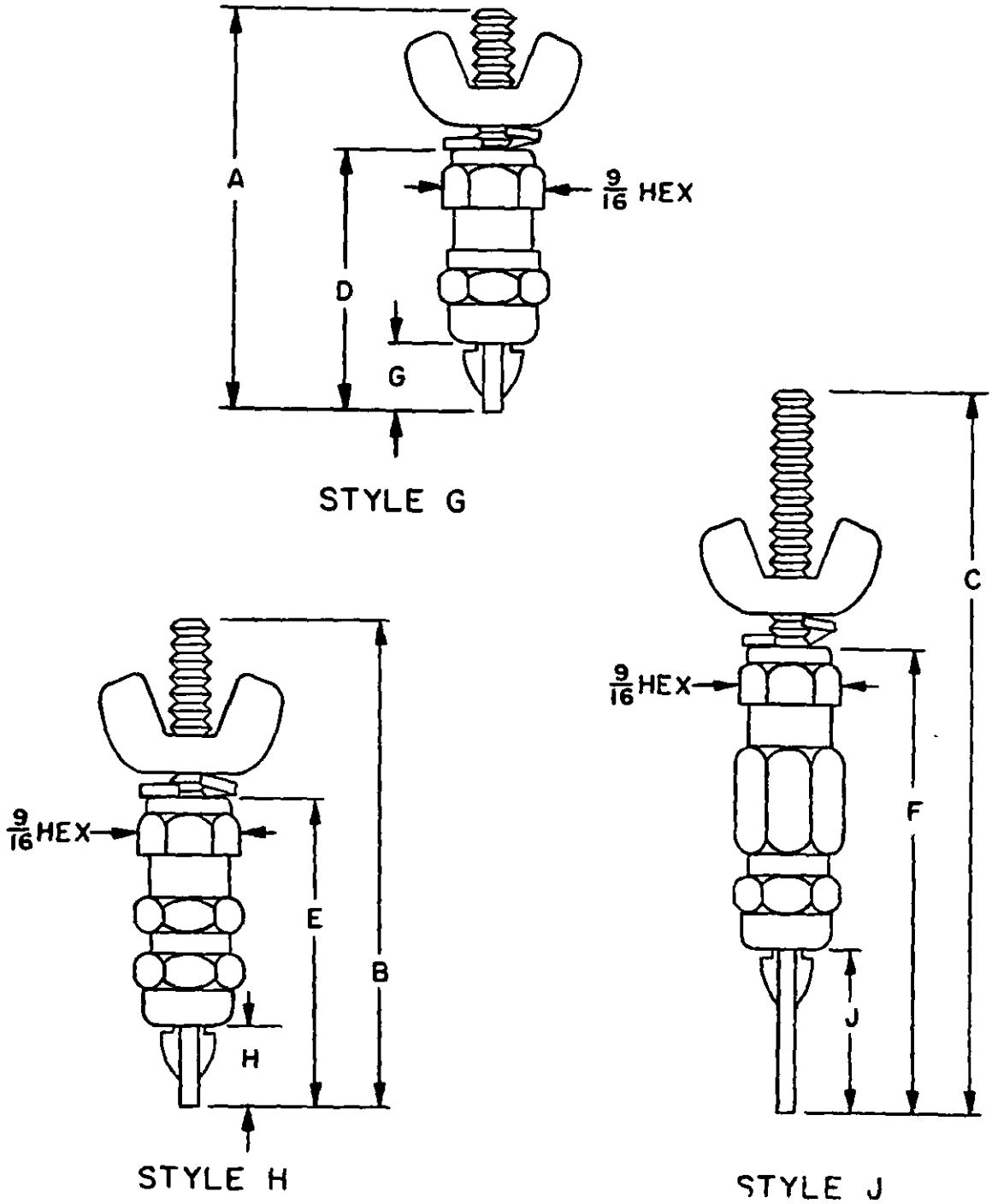


FIGURE 4. Type III, class 4, hole grip, hand-operated, screw type.

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TABLE V. *Detail requirements for type III, class 4, hole grip, hand-operated*

Nominal rivet size	Hole size standard to oversize	Dimensions in inches									Finish color
		A (max.)	B (max.)	C (max.)	D	E	F	G	H	J	
3/32	0.093 to 0.108	2-1/2	3-1/8	4-5/8	1-17/32	1-7/8	2-7/8	5/16	1/2	1	Cadmium
1/8	.125 to .140	↑	↑	↑	↑	↑	↑	↑	↑	↑	Copper
5/32	.156 to .171										Black
3/16	.187 to .202										Brass
7/32	.218 to .230										Cadmium
1/4	.250 to .265	↓	↓	↓	↓	↓	↓	↓	↓	↓	Copper
5/16	.312	2-1/2	3-1/8	4-5/8	1-17/32	1-7/8	2-7/8	5/16	1/2	1	Black

3.12 **Workmanship.** The holders shall be free from defects which adversely affect their appearance or which may affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* All sheet holders of the same type, class, and size offered for delivery at one time shall be considered a lot for purposes of acceptance, examination, and tests.

4.2.2 *Sampling for examination.* A random sample of sheet holders shall be taken from each lot for examination of visual and dimensional characteristics in accordance with MIL-STD-105 at inspection level I. The acceptable quality level (AQL) shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.2.3 *Sampling for lot acceptance tests.* A random sample of sheet holders shall be taken from each lot in accordance with MIL-STD-105 at inspection level S-3. The AQL shall be 1.0 percent defective. Each of the sample sheet holders shall be tested in accordance with 4.4.1, 4.4.2, and 4.4.3, as applicable, to verify conformance with this specification.

4.3 Examination.

4.3.1 *Visual and dimensional examination.* Each of the sample sheet holders taken in accordance with 4.2.2 shall be visually and dimensionally examined to determine conformance with this specification. Examination shall be conducted as specified in table VI. Any sheet holders in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective holders in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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TABLE VI. *Classification of defects for visual and dimensional examination*

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type, class, and size not as specified.
102	Incomplete, component parts missing or improperly assembled.
103	Materials not as specified.
104	Color code improper for size specified or color missing.
105	Dimensions not in accordance with the applicable table and figure.
106	Gripping range not as specified.
107	Gripping surface of jaw of improper material (type II, class 1 only).
Minor:	
201	Marking, manufacturer's name or trademark missing, incorrect, illegible, or not permanent.

4.4 Tests.

4.4.1 *Spring tension test.* The sheet holder shall be tested to determine compliance with the clamping force requirements specified in table I.

4.4.2 *Holding test.*

4.4.2.1 *Hole-grip holders.* The hole-grip types of sheet holders shall be tested for holding quality by inserting the holders into holes provided in sheet metal. The size of holes shall be the largest hole specified in table II for the holder being tested. With the holders inserted in the holes, they shall not come loose or fall out when struck a light blow with the hand at a sidewise position.

4.4.2.2 *Edge-grip holders.* The edge grip of sheet holders shall be tested for holding quality by clamping two pieces of metal 0.064 inch thick. A spring-pressure pull shall be applied to each holder in the same plane as that of the metal which is being held. The spring pressure required to remove the holder shall be not less than 10 pounds.

4.4.3 *Safety test for type I, class 2 holders.* The points of each holder selected for test shall be projected through a hole in a sheet of metal 3/16 inch thick and then held securely between the cutting jaws of a plier or nipper. With the holder in the clamped position, pressure shall be applied to the handles of the plier or nipper until the points are severed. The holder shall not fly away with a force that would injure the operator in any way and the holder shall remain intact.

4.4.4 *Possible test failures.* Possible test failures are defined as follows:

1. Spring-tension test:
   (a) Failure of springs to comply with requirements.
2. Holding test:
   (a) Sheet holders fail to hold properly when tested as specified.
3. Safety test (type I, class 2 only):
   (a) Holder flies away with a force that would injure the operator when tested as specified.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table VII. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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TABLE VII. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
	Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.2), the holders shall be preserved, packaged, and packed in accordance with MIL-H-15424. The levels of preservation and packaging shall be A or C and the level of packing A, B, or C, as specified (see 6.2).

5.1.1 As part of the requirements of section 5 of this specification, table VIII is considered as part of table I of MIL-H-15424.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The sheet holders covered by this specification are intended to hold sheet metal in place for drilling, trimming, and riveting during the fabrication and repair of aircraft, missiles, electronic chassis, and other sheet metal applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type and class required (see 1.2).
(*c*) Size required (see tables II through V).
(*d*) Whether 0 to 1/4, or 0 to 1/2 inch grip capacity is required (see table II and 3.9.1).
(*e*) Whether rubber or asbestos jaw surface is required on type II, class 1 holder (see 3.10.1).
(*f*) Selection of applicable levels of preservation, packaging, and packing required (see 5.1 and 5.1.1).

6.2.1 *For civil agency procurements.* When level B packaging is required for civil agency procurements, the level A requirements in 5.1 shall apply.

CUSTODIANS:
Army—Mobility Command (SMOFB-KX)
Navy—BuWeps (WP)
Air Force—MOAMA (69)

CIVIL INTEREST:
GSA-FSS
VA

Preparing activity:
Navy—Bureau of Naval Weapons (WP)

TABLE VIII.

	Column 2	Column 3	Column 4	Column 5	Column 6
Holders: Sheet metal, hole and edge grips	None	III	12	—	24

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2, of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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